CS404 — Big Data Analytics

ExposureIQ

Content Exposure Intelligence

Distributed Analytics of Global Torrent Activity

A batch + streaming research platform and
an analyst-facing company pilot

Research foundation	**Implemented product**
Public metadata, partitioned Parquet, Spark SQL and MLlib	Catalogue approval, timestamped checks, investigations, alerts and reports
Historical collection: 10 October 2026	Localhost, single-process ExposureIQ pilot

Team members

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Aligned to the supplied *Big Data Analytics* presentation

Metadata only. Tracker observations are not unique people, verified infringement, downloads or revenue loss. Historical findings are not live product measurements.

Executive summary

ExposureIQ turns a research pipeline over public torrent and entertainment metadata into a local workspace for content-protection analysts. Film distributors, streaming companies and game publishers can organise title catalogues, review timestamped tracker observations, assign investigations and export traceable records. The implemented product supports an analyst's review process; it does not establish infringement or perform enforcement.

The research foundation, *Distributed Analytics of Global Torrent Activity*, combines batch preparation and a streaming speed layer. Apache Spark cleans and joins RARBG, IMDb, TMDB, Steam and Wikipedia inputs into Parquet. Public UDP tracker counters flow through Kafka and Spark Structured Streaming to Parquet and PostgreSQL. Grafana supports the research dashboards; FastAPI and SQLite support the ExposureIQ company workspace.

The archived research run contains 1,153,632,847 valid Wikipedia pageview rows, 2,843,691 RARBG catalogue records, 184,661,759 rows across four IMDb staging tables, and 25,422,936 streamed observations. The latter were collected on 10 October 2026; the catalogue ends in May 2023. These time ranges must remain visible when interpreting observations.

Presentation query	**Saved finding**
Q1: Activity	Movie 1080p share rises from about 2% to 62% between 2008 and 2022.
Q2: Decay	Movie seeders follow a cross-sectional power-law association, approximately $\text{age}^{-0.73}$.
Q3: Release lag	Median first recorded cinema-to-listing lag is 9 days for 2020 and 2021 films in the selected sample.
Q4: Games	A logistic classifier of matched RARBG presence achieves test AUC 0.826.
Q5: Genres	Sci-Fi and Horror have positive listing-count residuals after popularity/year adjustment.
Q6: Interest	Weekly Wikipedia views and seeders have movie Spearman correlation 0.472.
Bonus: Intraday	A fixed-panel study finds a game index range of about 11.4 points, with smaller movie/TV variation.

The company pilot adds explicit match approval, saved portfolio checks, threshold rules, alerts, assigned cases, notes, exports and scheduled summary snapshots. Company records are organisation-scoped; the research catalogue and shared live collector are not private per-company streams.

All 26 entries in the archived validation file are labelled PASS. That is not equivalent to every artefact being complete: the JSONL backup is 1,111 records short, a timestamp-boundary check has a recorded discrepancy, and the original research Kafka topic was subsequently removed. This report preserves those caveats.

Evidence: `analysis/results/validation.csv;analysis/results/a1_*through7_*`

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1 Problem, scope and product evolution

1.1 Research problem

Torrent catalogues expose release metadata while public trackers report swarm counters. Analysing these systematically requires reconciling different identities, release dates, time ranges, formats and grains. Individual listings are not the same as titles, and tracker counters from overlapping swarms cannot be treated as independent audiences.

The project investigates how listing activity, release quality, release timing, title age, game characteristics, genre and public interest relate to the observed catalogue. Its scale includes over a billion pageview rows and tens of millions of tracker events. Spark executes locally on the project machine; this is distributed processing software in local mode, not evidence of a deployed multi-node cluster.

1.2 ExposureIQ's analyst-facing role

The current product is **ExposureIQ: Content Exposure Intelligence**. The academic subtitle remains useful because the presentation still introduces the underlying research before describing the product. ExposureIQ builds operational workflows around known titles and observations:

- Organise a company catalogue and explicitly approve proposed matches.
- Request known-hash checks and inspect coverage, timestamp and failures.
- Save portfolio observations and evaluate enabled threshold rules.
- Assign investigations, record notes and export reviewable evidence.
- Inspect a shared streaming pipeline and run bounded read-only queries.

The product does not continuously discover unknown new releases, verify content ownership, submit takedowns, or measure legal damages. A successful title match is a catalogue association, not an infringement determination.

1.3 Compatibility, not a second brand

User-facing names, report identity and exports use ExposureIQ. Existing `CWATCH_*` settings, `/api/contentwatch` routes, `X-CWatch-Request`, the `cwatch` PostgreSQL schema and the default `cwatch.sqlite3` database remain compatibility identifiers. Renaming those would require migrations for accounts, observations, checkpoints and clients; a documentation refresh must not silently break them.

The interface uses a dark, rounded visual theme with local assets. The System Health page has been removed; minimal and administrator-only health API endpoints remain. The public contact flow downloads an unsent enquiry rather than claiming server-side delivery.

Evidence: `portal/static/index.html;portal/static/contentwatch*.js;portal/server.py`

2 Company workflows and evidence semantics

2.1 From import to investigation

The demonstration sequence is **Import** → **Review** → **Approve** → **Check** → **Investigate** → **Report**. A company may also select individual titles directly from the shared research catalogue.

CSV onboarding accepts unique `title,category` headers and an optional `external_id`. The supported categories are movie, TV and game. Imports are limited to 200 rows and 1 MB, and validation failure imports nothing. Candidate matches remain pending until explicitly approved. An imported name absent from the source catalogue does not become monitorable merely because it was submitted.

Portfolio checks retain actual observation results. Ad-hoc search/check operations are not automatically persisted as portfolio history. Users must review timestamps, sampled hashes, responding trackers, missing coverage and explicit failures. Counts use the maximum tracker-reported seeders per record rather than a sum across overlapping trackers.

Record	Meaning and limitation
Catalogue entry	Company title and approval decision; not a tracker measurement.
Saved check	Timestamped known-hash observations and coverage; not complete discovery.
Rule / alert	Enabled threshold applied to a qualifying fresh check; not legal risk scoring.
Investigation	Assigned human review with state and notes; not automated enforcement.
Export	A traceable snapshot of records; generation time is not observation time.

2.2 Access, alerts and reporting

Administrators manage company access and operate the shared live collector. Analysts manage workflows and may run permitted live queries. Viewers and read-only API keys may inspect/export authorised records but cannot mutate company workflows. Private records are scoped to the current organisation.

An enabled title-specific rule should exist before a demonstration check. Unavailable or stale checks do not trigger rules, and an unresolved alert for the same rule/title suppresses duplicates. Optional email/webhook delivery has recorded states and errors; configured delivery is not proof that a notification arrived.

Reports include actual saved company summaries, catalogue CSV, observation CSV, complete check JSON and print-to-PDF workflows. Scheduled daily/weekly UTC summaries persist once per current period while the portal runs; they do not backfill missed periods. Signing out preserves records and does not stop the collector. Personal watchlists remain browser-local.

Evidence: `portal/accounts.py;portal/company.py;docs/EXPOSUREIQ_PILOT.md`

3 Architecture and separation of concerns

Batch: public source dumps / APIs
↓ acquisition and validation
Raw files → Spark cleaning → staging Parquet
↓ identity joins and research aggregates
Curated Parquet → Spark SQL / MLlib → saved results

Speed: public UDP tracker counts
↓ metadata scraper
Kafka → Spark Structured Streaming
↓ separate lake and serving queries
Parquet event history + PostgreSQL snapshots / aggregates / batch ledger

Serving: Grafana research dashboards + FastAPI ExposureIQ API
↓ company authentication and workflow operations
SQLite company state + browser workspace / exports

The batch layer provides historical context and title associations. The speed layer retains newly observed counters. PostgreSQL serves bounded queries over snapshots and aggregates, while Parquet retains larger event history. SQLite stores private operational records rather than duplicating the entire research lake.

Three collection contexts are distinct: the historical research run, the ExposureIQ product stream and the resettable classroom demo. The product uses topic `tracker.scrapes.cwatch`, schema `cwatch` and dedicated Parquet/checkpoint paths. Its starter creates three Kafka partitions. The presentation's six-partition configuration describes the research topic, not a universal product configuration.

Kafka, PostgreSQL, Grafana and optional scraper services are configured through Docker Compose. Spark runs on local cores. The original research environment was a 12-core machine with approximately 15 GB RAM; the report does not claim cluster benchmarks or horizontal scalability measurements.

The product's **Query** editor executes PostgreSQL over `live_latest`, `live_minutes` and `live_batches`. Research examples in `demo.sql` use Spark SQL. These engines, views and result grains must not be conflated.

Evidence: `infra/docker-compose.yml;streaming/swarm_stream.py;portal/live.py`

4 Data sources, time coverage and identities

Source / prepared surface	Saved rows	Coverage
Wikipedia valid hourly pageviews	1,153,632,847	1–7 Sep 2026
IMDb: four staging tables	184,661,759	Research snapshot
RARBG catalogue	2,843,691	2007–May 2023
Steam games after deduplication	125,854	Oct 2026 snapshot
TMDB title records	13,462	Selected titles
TMDB release-date records	212,933	Selected titles
Historical stream observations	25,422,936	10 Oct 2026
Full-catalogue tracker scan	1,385,068	10 Oct 2026

Counts describe distinct surfaces and are not additive estimates of unique people or titles. IMDb's total refers to basics, alternative titles, principals and episodes. The roughly 1.38-billion-row research description is a scale summary, not a deduplicated entity count.

The presentation contains an approximate 1.5-billion pageview statement; its source list and archived validation support **1,153,632,847 valid pageview rows**. This report uses the latter rather than propagating inconsistent rounding.

4.1 Identity resolution

A torrent's `infohash` identifies one release. IMDb IDs join film/TV metadata to release dates; Steam app IDs identify games. Multiple release hashes can map to one title. Wikipedia entities are matched by page title and confidence, using English desktop/mobile views for the interest study. The product's `title_key` identifies a title, not a torrent.

Game name normalisation associates about 79% of relevant game torrents with Steam entries. The presentation reports 34 correct matches in a 35-item manual check; this is a small spot check, not a population-wide accuracy guarantee. Unmatched games affect model outcome labels.

4.2 Dates are part of the evidence

The RARBG catalogue stops in May 2023. Tracker measurements were collected in October 2026, while the Wikipedia week is in September 2026. A chart combining these sources is not a contemporaneous user-behaviour panel. On-demand product checks carry their own actual timestamps.

Evidence: `analysis/results/validation.csv;dataset/README.md;docs/schema.md`

5 Preparation, streaming and reproducibility

5.1 Batch preparation

Raw inputs are retained separately from cleaned staging and joined curated tables. Spark writes compressed, partitioned Parquet for the research lake. Release-name parsing extracts title, year, season/episode, resolution, codec and source. The presentation records `guessit` parsing through `mapInPandas` on 12 workers, taking about 60 minutes for the RARBG catalogue; this is an archived run observation, not a current benchmark.

Cleaning includes infohash deduplication, normalisation of catalogue categories, tolerant casting, malformed-line handling and explicit Steam CSV schema repair. The Steam header has 39 names for 40 values because Discount and DLC count were merged. Wikipedia loses 93,017 malformed lines out of 1,153,725,864 raw lines (about 0.0081%); these are excluded, not silently treated as valid pageviews.

Exact IMDb identifiers support film/TV joins; normalised names support Steam associations. Broadcast joins reduce large pageview scans against small title lookup tables. The presentation reports about one minute for its selected Wikipedia title join; it does not establish end-to-end pipeline latency.

5.2 Stream processing and replay

Scrapers request public tracker counters using BEP 15 UDP scrape messages. They do not request peer lists. Kafka JSON includes hash, tracker, seeders, leechers, completed count and observation timestamp. Spark parses events and excludes rows lacking a usable hash or timestamp.

One query appends Parquet; another applies serving micro-batches. The serving transaction inserts a query/batch ledger record and updates minute summaries and latest observations together. A replayed ledger key is skipped, avoiding duplicate serving application for that batch. Latest observations are keyed by infohash/tracker and updated using timestamp order.

The two sinks have separate progress/checkpoint state. Transactional serving replay protection is not a blanket claim of atomicity across Kafka, Parquet, PostgreSQL and JSONL. Kafka retention, checkpoint loss, input drops and forced shutdown still need operational handling.

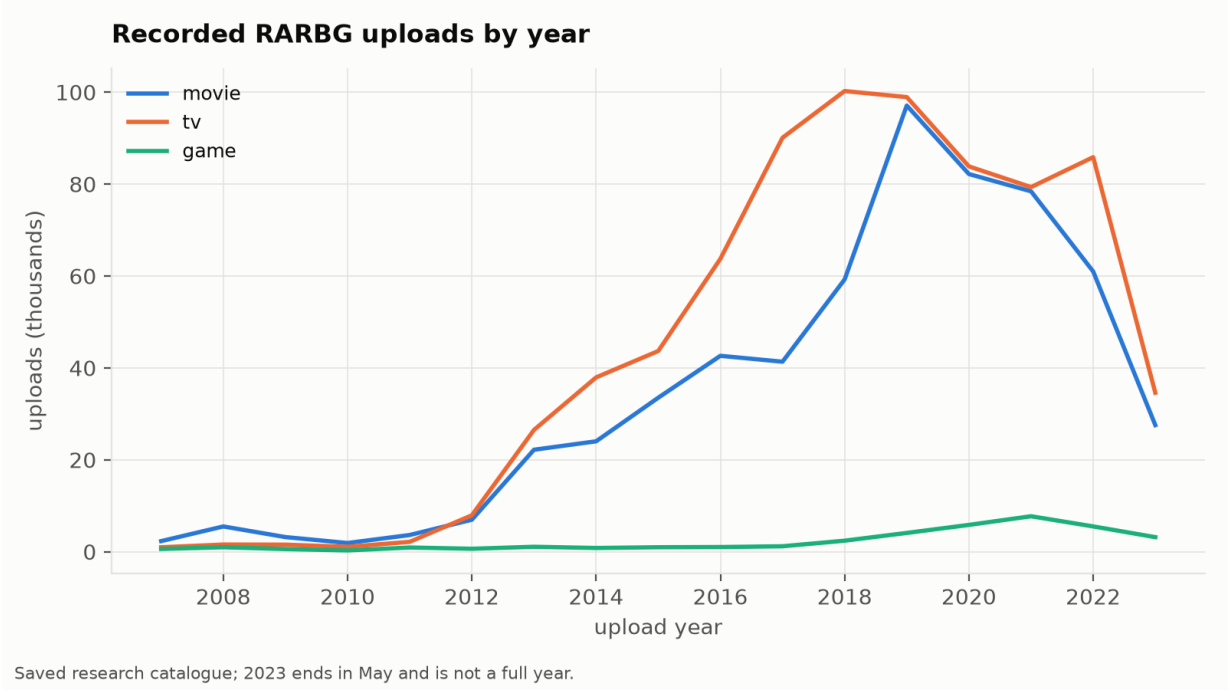
5.3 A lightweight report rebuild

The replacement charts read tracked result CSVs and reuse the repository's plot style. Rebuilding this PDF does not rerun Spark, acquire datasets, contact trackers, change results or read company records. It therefore reproduces the presentation of saved evidence, not the original experiment. Full research validation requires the prepared lake and appropriate services.

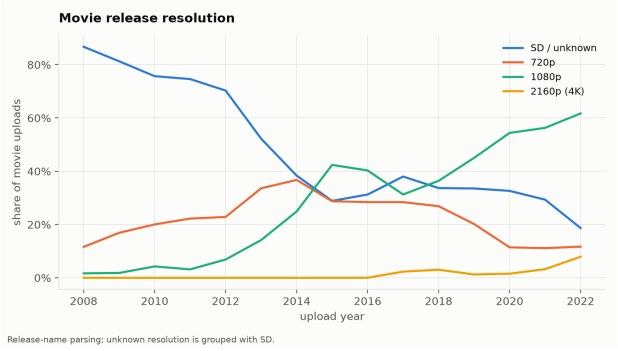
Evidence: `etl/;ingest/;streaming/swarm_stream.py;analysis/render_saved.py`

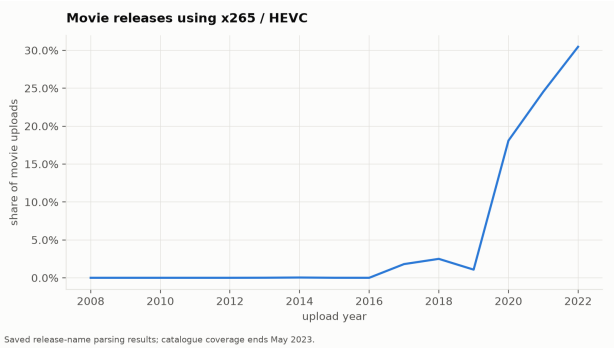
6 Q1: Activity and release quality (A1)

Question. How have upload volume and release quality changed by category? The study groups catalogue records by upload year, then derives movie resolution and codec shares from parsed release names.



Upload counts describe listings, not unique titles or downloads.





TV uploads rise from 1,068 in 2010 to roughly 80,000–100,000 per year over 2018–2022. Movie 1080p share increases from about 2% in 2008 to 62% in 2022; x265/HEVC share increases from about 1% in 2019 to 30% in 2022. Different formats and repacks can represent the same underlying title.

Analyst use. These trends provide historical context for release formats and catalogue composition. They do not establish which formats a current rights holder should prioritise without current, licensed evidence.

Limits. RARBG is one catalogue; removed releases and site-specific listing practices affect counts. 2023 ends in May and must not be compared as a full year. Parsed labels can be missing or ambiguous.

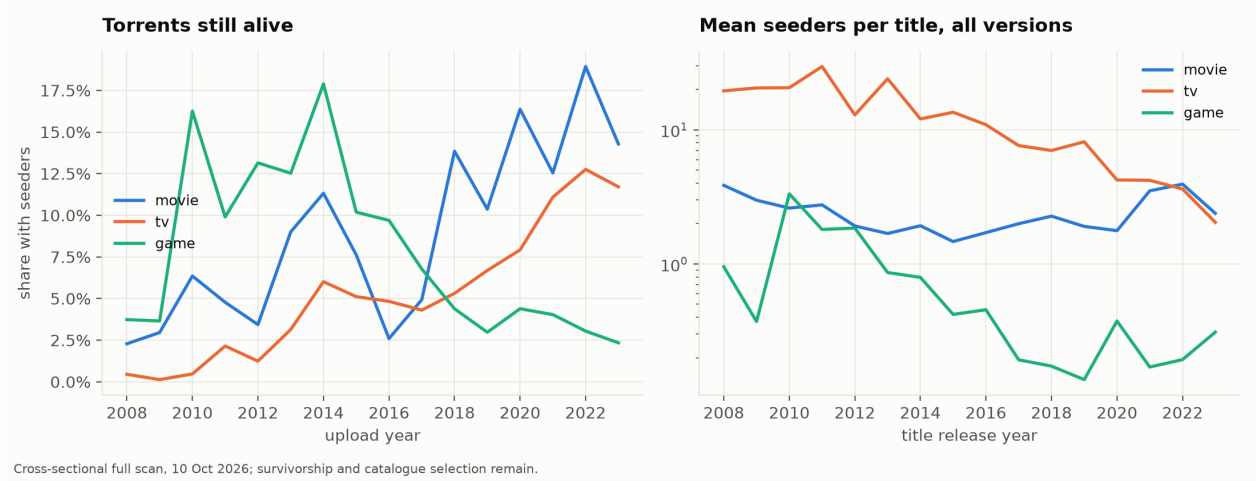
Evidence: `analysis/results/a1_uploads_by_category_year.csv`

Evidence: `analysis/results/a1_movie_resolution_share.csv;a1_movie_x265_share.csv`

7 Q2: Popularity decay and persistence (A3)

Question. How does observed swarm size vary with title age? The research aggregates seeders across versions per title, then fits Poisson generalised linear models for exponential and power-law forms. The selected fit population uses title release years 2012–2023.

$$\log E[S \mid a] = \beta_0 + \beta_1 a \quad \text{or} \quad \log E[S \mid a] = \beta_0 + \beta_1 \log a.$$



Cross-sectional scan: title release cohorts are not longitudinal trajectories.

Category	Titles	Best saved fit by AIC
Movie	34,632	Power law: coefficient −0.733; AIC 401,682.94
TV	10,301	Power law: coefficient +1.647; no negative decay association
Game	10,871	Exponential: coefficient +0.157; positive age association

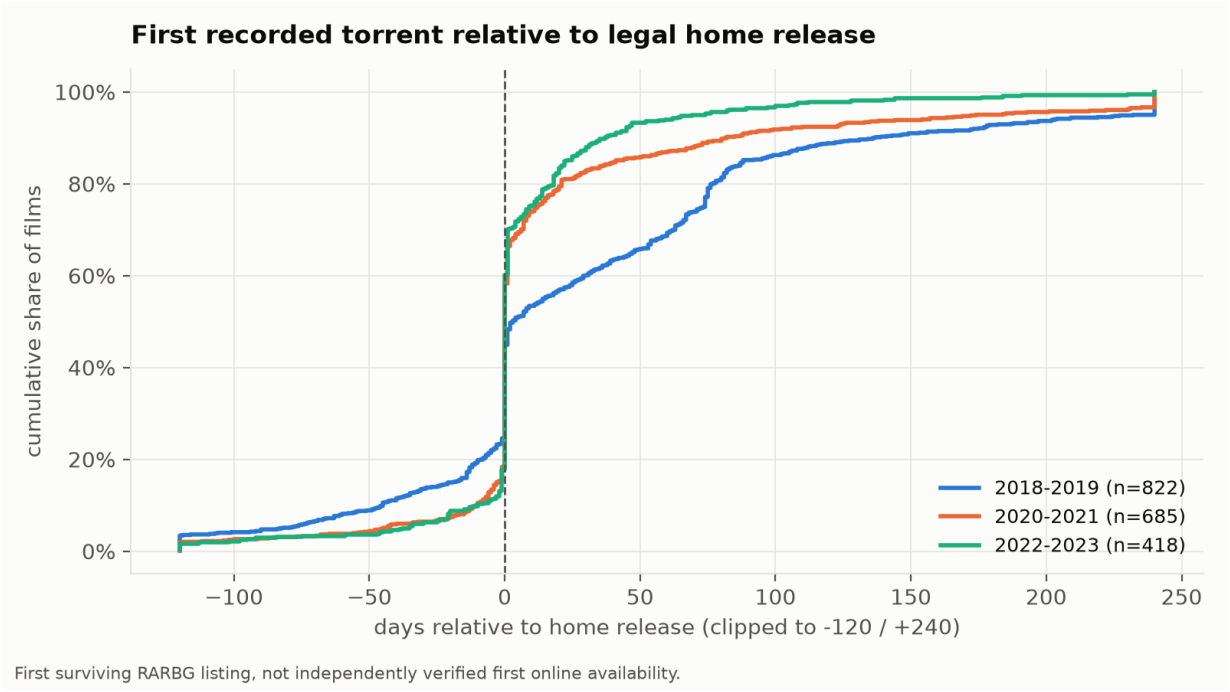
For films, the exponential alternative has AIC 402,250.00, about 567 points higher. The preferred association is approximately $S \propto a^{-0.73}$. The positive TV/game fits must not be described as negative decay; continuing series and persistent older games may contribute.

Analyst use and limits. The result motivates attention to long-lived known titles. It does not predict an individual title's future trajectory. Survivorship, missing old listings, title matching and version splitting remain. The presentation's 9.0% alive-torrent figure describes the archived full-scan cross-section, not a current product success rate.

Evidence: `analysis/results/a3_title_decay_fits.csv;a3_title_decay_by_release_year.csv`

8 Q3: Release-to-listing lag (A2)

Question. How soon after cinema and legal home release do films appear in the recorded catalogue? The first surviving RARBG listing per film is compared with TMDB release dates. The selected movie window is 2018 through 31 March 2023; older release cohorts are too incomplete to support the same interpretation.



Zero marks legal home release; a negative lag is an earlier recorded listing.

Year	Films	Median days	Interpretation
2018	429	76	Cinema to first surviving listing
2019	411	75	Cinema to first surviving listing
2020	322	9	Shorter observed lag
2021	367	9	Shorter observed lag
2022	363	18	Selected catalogue sample
2023	61	19	Partial release-year coverage

The presentation reports 66–70% of the 2020–2023 cohort listed by the day after legal home release, and a one-day median from TV series premiere to the first recorded series torrent. “Home” means the earliest qualifying digital or physical date, not necessarily a single worldwide launch date.

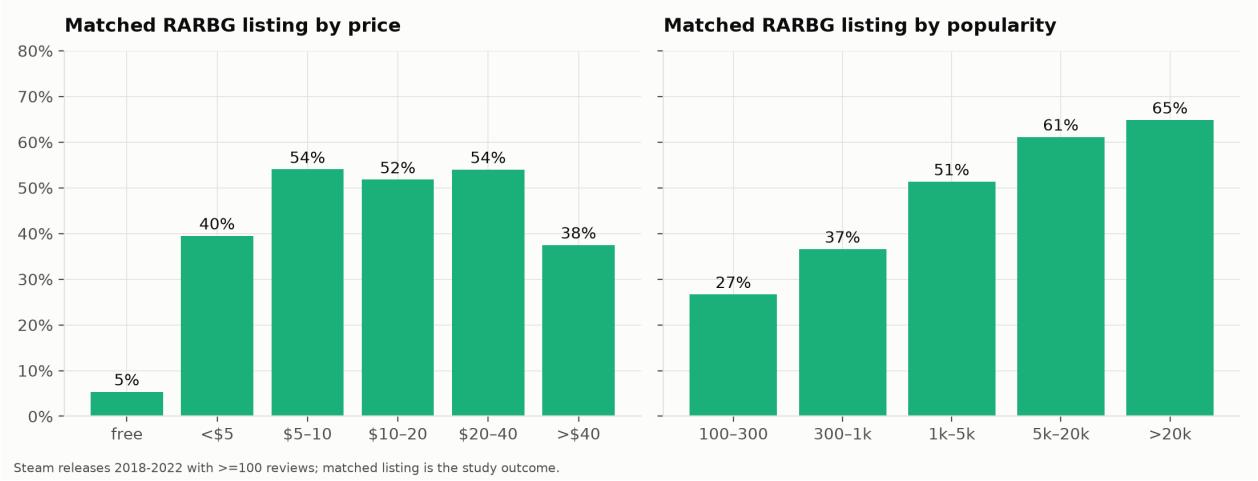
Analyst use. Release windows can guide when analysts review known titles. They are not a guarantee of automated early discovery.

Limits. First surviving catalogue listing is not independently verified first online availability. Country-specific windows, replacements, missing listings and incomplete matching can change lags. The historical pattern is not evidence that every present-day release leaks within one day.

Evidence: `analysis/results/a2_movie_first_upload.csv;a2_movie_leak_by_year.csv;a2_tv_first_upload.csv`

9 Q4: Game characteristics and catalogue presence (A5)

Question. How do price, popularity and game type relate to the presence of a matched RARBG listing? The population is Steam games released 2018–2022 with at least 100 reviews. The study label "pirated" means at least one matched RARBG game torrent; it does not verify every underlying file or legal status.



Descriptive proportions by price and review-count band; these are not causal price effects.

The logistic model uses standardised features, an 80/20 split with seed 42, and regularisation. Test AUC is **0.8263**, rounded to 0.83 in the presentation. Odds ratios are per one standard deviation of each feature, not a per-dollar effect.

Feature	**Approximate odds ratio / SD**
Popularity (log reviews)	1.84
Single-player	1.79
Price (log transformed)	1.55
Massively multiplayer	0.77
Casual	0.71
Free-to-play	0.54
Early Access	0.51

Free-to-play titles have about 5.4% matched presence; $5–40 bands have roughly 52–54%. The above-$40 band is 37.5% but contains only **32 games**, so the apparent decline deserves caution.

Analyst use and limits. These associations support contextual prioritisation, not individual guilt or causal pricing recommendations. Unmatched torrents create incomplete outcome labels. The split evaluates held-out catalogue labels, not unseen future markets or production alerts.

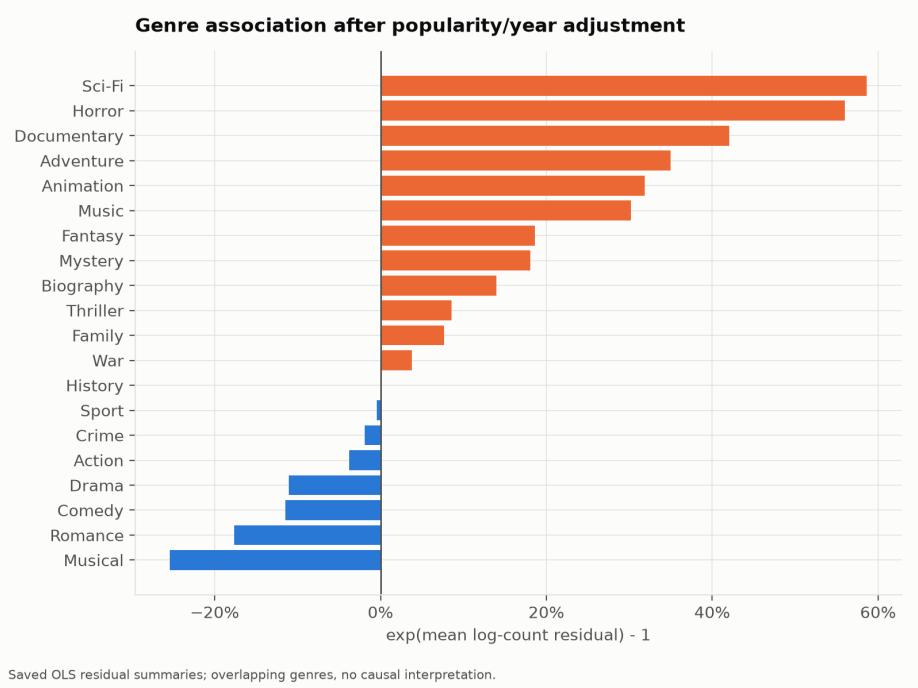
Evidence: `analysis/results/a5_logit_pirated.csv;a5_pirated_by_band.csv;analysis/a5_games.py`

10 Q5: Genre associations beyond popularity (A6)

Question. Which film genres have more recorded versions than their popularity and release year would predict? The study fits an OLS model of log listing counts and summarises residuals by genre:

$$\log(1 + \text{torrents}) \sim \log(\text{IMDb votes}) + \text{release-year indicators}.$$

The displayed effect is exp(mean residual) − 1.



Residual-based associations; genres overlap and are not randomised treatments.

The saved result gives approximately **Sci-Fi +59%**, **Horror +56%**, **Romance** −18% and **Comedy** −12%. The model's saved R^2 is 0.1933: most variation remains unexplained. A residual effect is not the fraction of a genre's audience engaged in infringement.

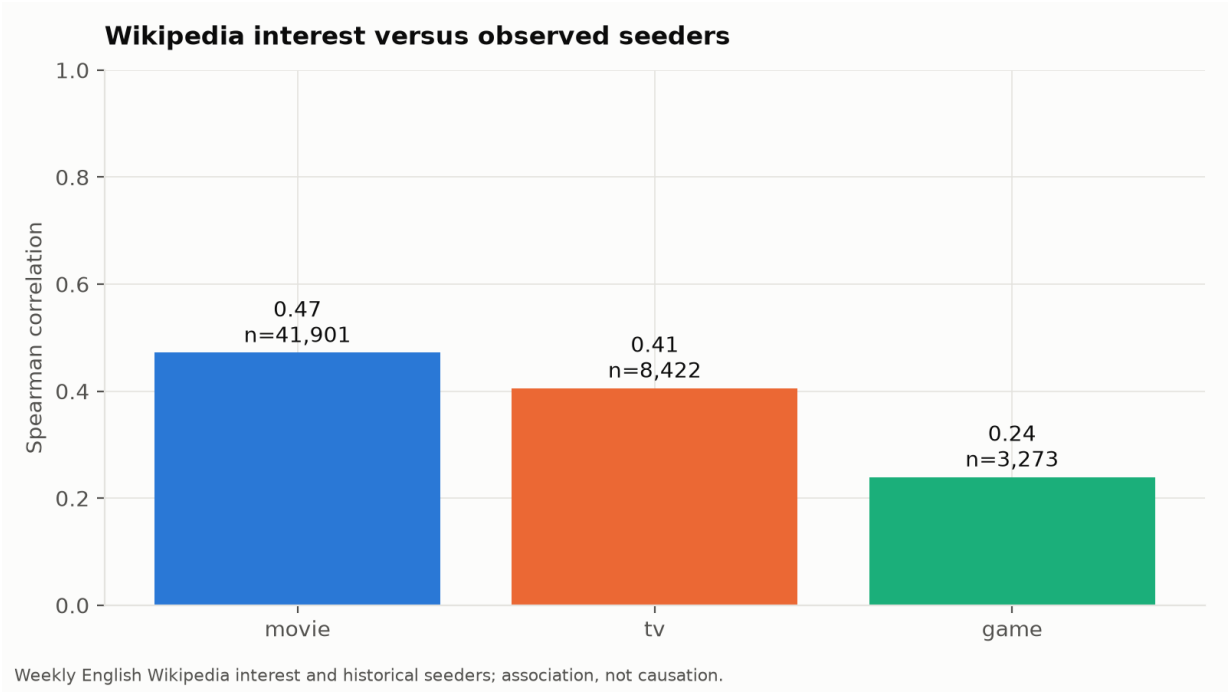
Analyst use. Genre can provide background context for a catalogue review queue when combined with actual company titles, current observations and human judgement. ExposureIQ does not turn these research coefficients into an automatically validated legal risk score.

Limits. IMDb votes are a popularity proxy and may correlate with unmeasured factors. Genre memberships overlap, catalogue selection is incomplete and residual summaries do not establish causality. Multiple versions and release practices can affect the count outcome.

Evidence: `analysis/results/a6_genre_excess_piracy.csv;a6_model.csv;analysis/a6_genres.py`

11 Q6: Public interest and observed seeders (A4)

Question. Does Wikipedia interest track the observed torrent activity of matched titles? The study compares weekly English Wikipedia views with seeders using Spearman rank correlation.



Rebuilt correlation summary from the saved results, not a new scatter-plot experiment.

Category	Matched titles	Spearman ρ	Reading
Movie	41,901	0.4718	Moderate positive rank association
TV	8,422	0.4056	Moderate positive rank association
Game	3,273	0.2394	Weaker positive rank association

The three samples sum to 53,596 titles. Correlation is far from one, so public interest is only a partial proxy for observed swarm activity. It is not a conversion rate from pageviews to downloads.

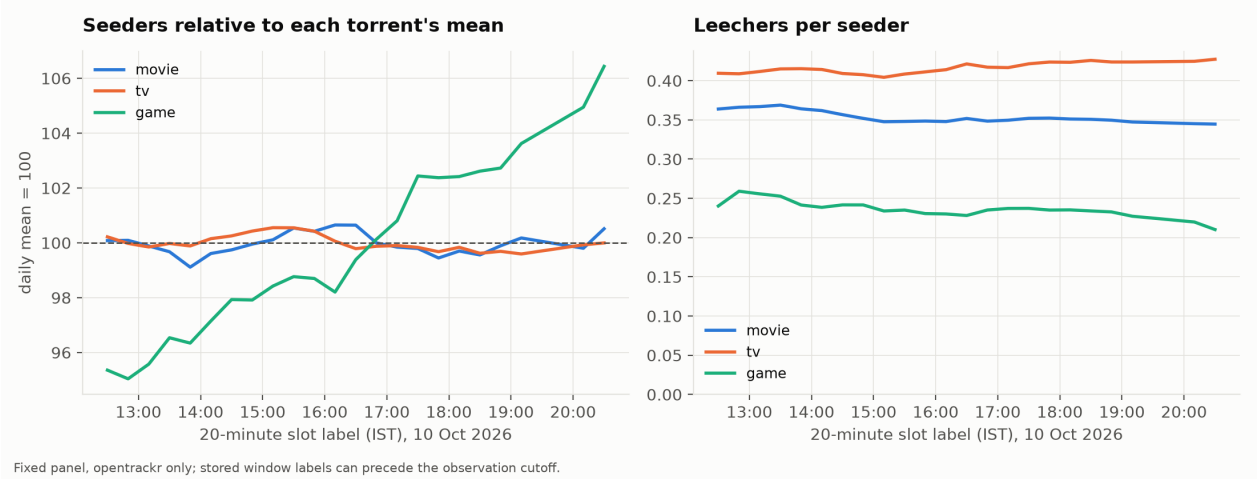
Analyst use. Interest metadata can contextualise why a title may attract attention, but fresh timestamped checks are needed to describe current tracker responses. The product should not label a title as currently exposed solely because it is popular.

Limits. Wikipedia matching can be ambiguous, and English desktop/mobile views are not global demand. The interest week and the tracker collection are not simultaneous. Both signals may reflect other causes, including marketing or legal release timing.

Evidence: `analysis/results/a4_spearman.csv;analysis/a4_wiki_interest.py`

12 Bonus: Swarms during one day (A7)

Question. How does observed swarm size vary within the research collection day? The design uses opentrackr only, a fixed panel and 20-minute windows to reduce tracker-outage and changing-sample artefacts. Each torrent is indexed relative to its own daily mean before averaging.



The chart retains stored IST window labels; a window start may precede the observation cutoff.

Category	Min index	Max index	Mean leechers/seeder
Movie	99.11	100.65	0.354
TV	99.59	100.55	0.416
Game	95.04	106.43	0.236

The index range is 1.54 points for movies, 0.96 for TV and 11.40 for games. The game's range is not automatically an 11.4% increase from its first observation. The presentation's "+11%" is a rounded description of this intraday pattern, not a demonstrated repeatable daily cycle.

The observation cutoff is 12:40–20:50 IST on 10 October 2026. Spark window alignment can produce an earlier displayed slot label; labels must not be mistaken for observations outside the filter. Panel eligibility requires observations in at least 90% of slots and positive median seeders; slots need at least 80% panel coverage.

Analyst use and limits. The result illustrates why sampling design matters for dashboards. A one-day, selected alive-torrent panel cannot establish global time-zone behaviour or a general daily seasonality model. Tracker counters remain partial observations, not unique users.

Evidence: `analysis/results/a7_panel_by_slot.csv;a7_summary.csv;analysis/a7_swarm_day.py`

13 Validation, performance and evidence quality

13.1 What the archived checks establish

The saved file has 26 PASS-labelled entries, including source/staging reconciliation, hash uniqueness, 168 Wikipedia hourly files, count sanity, cross-tracker agreement and selected parsing/release-date spot checks. They are recorded results from the original experiment, not freshly executed checks during this report rebuild.

Archived measurement	Result and scope
Kafka / serving ledger / Parquet	Each records 25,422,936 observations for the historical run.
Duplicate serving application	Zero recorded duplicate micro-batches; 603 committed batches.
Tracker agreement	Spearman 0.736 on selected cross-tracker pairs.
Full scan versus near-time live scrape	Spearman 0.977 on selected same-tracker pairs.
Presentation peak throughput	Approximately 3.56 million rows/hour; not an end-to-end latency SLA.
Total research collection	About 10.4 hours; 25.4 million observations.

13.2 Caveats that PASS labels do not erase

JSONL backup shortfall. The backup has 25,421,825 lines, which is 1,111 fewer than Kafka/Parquet. Its note attributes the difference to an unflushed write buffer when scraper containers were force-stopped. Thus "zero lost" in the presentation applies to the reconciled primary pipeline, not every secondary backup.

Timestamp envelope. The archived boundary check lists an expected 05:06–15:31 UTC envelope and an observed 05:05–15:30 UTC range, yet is marked PASS. This report records the discrepancy rather than claiming an exactly satisfied boundary test.

Removed source topic. Kafka totals reference an offset snapshot recorded on 10 October 2026; the original topic was later removed. The tracked snapshot supports historical provenance but cannot substitute for querying a still-existing original topic.

Product versus research testing. Unit tests cover company and portal behaviour separately. Saved research validation is not a substitute for verifying current authentication, isolation, live startup, failure handling and exports. A report build is not a production certification.

Evidence: `analysis/results/validation.csv;kafka_offsets_snapshot.json;portal/test_*.py`

14 Operating, demonstrating and preserving the project

14.1 Start the existing pilot

With prepared dependencies and research data, run from the repository root:

```
uv run python -m portal.server
# http://localhost:8800
```

Create a company or sign in. Do not reacquire datasets, rerun ETL or reset a classroom stream just to open the website. The administrator can start known-hash monitoring with 100, 500 or 2,000 targets. Docker, Java and populated historical PostgreSQL serving data are required for shared streaming.

The product's query connection is a dedicated non-superuser read-only PostgreSQL login with a five-second statement timeout, one-second lock timeout and 200-row cap. Include observation-time filters when asking about fresh records. Retained snapshots remain queryable after collection stops.

14.2 Rebuild the report and run checks

```
uv run python -m analysis.render_saved
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  -cd report/ExposureIQ_Report.tex
uv run python -m unittest discover -s portal -p 'test_*.py'
```

The supplied presentation is preserved separately under `showcase/`. The report's editable LaTeX, rendered PDF, charts and saved research CSVs are suitable tracked artefacts. Company records are not.

14.3 Backup before reclaiming disk space

A Git push preserves tracked repository content only. It does not preserve the ignored data lake, the company database, credentials, the local ZIP, browser watchlists or Docker-backed Kafka/PostgreSQL/Grafana state. Checkpoints preserve processing progress, not the full observation history.

Use the provided SQLite backup API rather than copying only an active database main file:

```
uv run python -m portal.manage backup \
  --file /safe/private/location/exposureiq.sqlite3
```

Stop the portal before restore. The restore tool invalidates sessions, revokes API keys and marks invitations used. Keep private backups outside the public checkout. Neither chart generation nor PDF compilation needs dataset deletion or cache cleanup.

Evidence: `portal/manage.py;docs/EXPOSUREIQ_PILOT.md;report/README.md`

15 Limitations, ethics and conclusion

15.1 Measurement limits

RARBG represents one catalogue ending in May 2023, not the whole torrent ecosystem. Public trackers observe only part of each swarm. Counts can overlap across trackers and releases and do not identify people. Unknown hashes are outside the current product's monitoring scope. Matching errors, incomplete release dates and removed old listings affect all downstream conclusions.

The decay study is cross-sectional, not a repeated observation of every title over its lifespan. The intraday study covers one selected panel on one day. Game labels depend on incomplete name matching. Wikipedia interest is language- and period-specific. Regression and correlation findings describe associations, not causal effects.

15.2 Privacy and responsible use

The implementation analyses metadata and tracker counters, not copyrighted media content. Scrapers do not request peer lists or store peer IP addresses. Company records and credentials remain private and excluded from source control. Analysts must retain timestamps, coverage and limitations when sharing observations; an alert is a review prompt, not a finding against an individual.

Public accessibility does not remove source licence obligations. In particular, IMDb inputs are non-commercial. Any commercial deployment requires review and replacement/licensing of unsuitable sources. Source terms and lawful use must be evaluated before expanding collection.

15.3 Pilot boundaries and future work

ExposureIQ currently runs as a single-process localhost pilot. Shared stream ownership and reader credentials are managed in process memory. There is no billing, MFA, account recovery or email verification service. Internet deployment requires HTTPS, stronger operational safeguards, recovery procedures and validation under the intended tenant/deployment model. No production-readiness or enforcement claim is made.

Potential future work includes licensed contemporary catalogue sources, evaluated discovery mechanisms, longer longitudinal studies, robust deployment/recovery and better matching evaluation. These are future requirements, not implemented product features.

15.4 Conclusion

The project connects large-scale historical research with a concrete analyst workflow. ExposureIQ makes title decisions, observations and human review more traceable while preserving the original six research questions and intraday streaming study. Its credibility depends on separating historical findings from fresh checks, public metadata from private company state, and measured associations from legal or causal conclusions.

16 Evidence map and references

16.1 Presentation order and implementation

Deck	Module	Tracked result evidence
Q1 Activity	A1	Upload counts, resolution shares, codec shares
Q2 Decay	A3	Release-cohort aggregates, model coefficients and AIC
Q3 Release lag	A2	First listings, yearly summaries, TV lag
Q4 Games	A5	Standardised odds ratios, AUC, price/review bands
Q5 Genres	A6	OLS coefficients and genre residual summaries
Q6 Interest	A4	Category sample sizes and Spearman correlations
Bonus Intraday	A7	Fixed-panel slot observations and category summaries

Research source and CSVs are under `analysis/` and `analysis/results/`. The product implementation is under `portal/`; stream processing is under `streaming/`. The data/identity reference and operator guide are under `docs/`. The report's build instructions are `report/README.md`.

16.2 Supplied presentation

Big Data Analytics, Shubham Sharma and Ajay Singh, CS404. User-supplied October 2026 PDF, preserved as `showcase/Big Data Analytics.pdf`. The presentation retains the academic project title and introduces ExposureIQ as the intended analyst workspace. Its research charts and rounded claims are interpreted alongside saved result tables, not as evidence of new measurements.

16.3 Prior work cited in the presentation

1. Pouwelse et al. (2005), BitTorrent measurement, availability and flash crowds, IPTPS. Presentation reference: https://pdos.csail.mit.edu/archive/6.824-2007/papers/pouwelse-btmeasure.pdf.

2. Guo et al. (2005), torrent popularity and peer-to-peer measurement, IMC. Presentation reference: https://static.usenix.org/event/imc05/tech/full_papers/guo/guo.pdf.

3. Meulpolder et al. (2010), *Public and Private BitTorrent Communities: A Measurement Study*, IPTPS. https://www.usenix.org/conference/iptps-10/public-and-private-bittorrent-communities-measurement-study.

4. Zhang et al. (2011), public BitTorrent ecosystem measurement, IEEE TPDS. Presentation reference: https://cse.engineering.nyu.edu/~ross/papers/PublicEcosystem.pdf.

16.4 Source and protocol references

IMDb non-commercial datasets: https://datasets.imdbws.com/.
Wikimedia hourly pageviews: https://dumps.wikimedia.org/other/pageviews/.
TMDB developer API: https://developer.themoviedb.org/.
Steam dataset (fronkongames): https://www.kaggle.com/datasets/fronkongames/steam-games-dataset.
BEP 15 UDP tracker protocol: https://www.bittorrent.org/beps/bep_0015.html.